

EDUCATION

- **University of Wisconsin-Madison** Madison, WI
Ph.D. - Mathematics (Minor: Computer Sciences), Advisor: Nan Chen Jul 2023 – present
- **Nanjing University** Nanjing, China
M.S. - Atmospheric Science, Advisor: Lili Lei Sep 2020 – Jun 2023
B.S. - Atmospheric Science Sep 2016 – Jun 2020

RESEARCH INTERESTS

My research lies at the interface of applied and computational mathematics, machine learning, and geophysics. I am interested in the analysis, modeling, and filtering of complex dynamical systems, which draws on data assimilation, uncertainty quantification, stochastic differential equations, and information and learning theory. My current focus is developing accurate and efficient data assimilation algorithms in concert with scientific machine learning.

PUBLICATIONS

- **Zhongrui Wang**, Rebecca Willett, Nan Chen, and Yinling Zhang. “A Regime-Dependent Multi-Model Framework for Data Assimilation and Prediction”, *Journal of Advances in Modeling Earth Systems* (under review), 2026
- **Zhongrui Wang**, Chuanqi Chen, Jin-Long Wu, and Nan Chen. “A Lagrangian Conditional Gaussian Koopman Network for Data Assimilation and Prediction”, *Journal of Computational Physics* (under review), 2026
- Chuanqi Chen, **Zhongrui Wang**, Nan Chen, and Jin-Long Wu. “Modeling partially observed nonlinear dynamical systems and efficient data assimilation via discrete-time conditional Gaussian Koopman network”, *Computer Methods in Applied Mechanics and Engineering* 445, 118189, 2025.
- **Zhongrui Wang**, Nan Chen, and Di Qi. “A Nonlinear Data Assimilation Algorithm with Closed-Form Approximations for Multilayer Flow Fields”, *Monthly Weather Review* 153, 12, 2889–2912, 2025.
- **Zhongrui Wang**, Haohao Sun, Lili Lei, Zhe-Min Tan, and Yi Zhang. “The importance of data assimilation components for initial conditions and subsequent error growth”, *Science China Earth Sciences*, 2024.
- **Zhongrui Wang**, Lili Lei, Jeffrey L. Anderson, Zhe-Min Tan, and Yi Zhang. “Convolutional Neural Network–Based Adaptive Localization for an Ensemble Kalman Filter”, *Journal of Advances in Modeling Earth Systems* 15, 10, e2023MS003642, 2023.
- Lili Lei, **Zhongrui Wang**, and Zhe-Min Tan. “Integrated Hybrid Data Assimilation for an Ensemble Kalman Filter”, *Monthly Weather Review* 149, 12, 4091–4105, 2021.
- Shibao Wang, Yun Ma, **Zhongrui Wang**, Lei Wang, Xuguang Chi, Aijun Ding, Mingzhi Yao, Yunpeng Li, Qilin Li, Mengxian Wu, Ling Zhang, Yongle Xiao, and Yanxu Zhang. “Mobile monitoring of urban air quality at high spatial resolution by low-cost sensors: impacts of COVID-19 pandemic lockdown”, *Atmospheric Chemistry and Physics* 21, 7199–7215, 2021.
- Yanxu Zhang, Xingpei Ye, Shibao Wang, Xiaojing He, Lingyao Dong, Ning Zhang, Haikun Wang, **Zhongrui Wang**, Yun Ma, Lei Wang, Xuguang Chi, Aijun Ding, Mingzhi Yao, Yunpeng Li, Qilin Li, Ling Zhang, and Yongle Xiao. “Large-eddy simulation of traffic-related air pollution at a very high resolution in a mega-city: evaluation against mobile sensors and insights for influencing factors”, *Atmospheric Chemistry and Physics* 21, 2917–2929, 2021.

CONFERENCES AND SEMINARS

- **Talk** (upcoming): 8th WMO Symposium on Data Assimilation & 12th International Symposium on Data Assimilation (ISDA), Nanjing, Aug-Sep 2026
- **Poster**: AI+Science Summer School, University of Chicago, Chicago, Jun 2026
- **Talk**: SIAM Conference on Uncertainty Quantification (UQ26), Minneapolis, Mar 2026
- **Talk**: Scientific Machine Learning: Theory, Algorithms, and Applications, Purdue University, Sep 2025
- **Poster**: IMSI Digital Twins: Mathematical and Statistical Foundations and Complex Applications, Chicago, Sep 2025
- **Poster**: Data Meets Dynamics: Workshop on Data Assimilation for Complex Systems and Applications, Madison, Aug 2025
- **Talk**: Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Jul-Aug 2025
- **Talk**: SIAM Conference on Applications of Dynamical Systems (DS25), minisymposium on nonlinear and stochastic dynamics in weather and climate science, Denver, May 2025
- **Talk**: The Institute for Foundations of Data Science (IFDS) ideas forum, Madison, May 2025
- **Talk**: ISDA Online: Nonlinear Data Assimilation, Online, May 2025
- **Poster**: The American Geophysical Union (AGU) Annual Meeting, Washington, D.C., Dec 2024
- **Talk**: 102nd American Meteorological Society (AMS) Annual Meeting, 26th Conference on Integrated Observing and Assimilation Systems for the Atmosphere, Oceans, and Land Surface (IOAS-AOLS), Jan 2022

ACADEMIC SERVICE

- Reviewer: Physica D: Nonlinear Phenomena, 2026
- Reviewer: npj Artificial Intelligence, 2025

TEACHING

- **Teaching Assistant** University of Wisconsin-Madison
Math 211: Survey of Calculus I Fall 2025, Spring 2026
- **Undergraduates Mentoring** University of Wisconsin-Madison
Madison Experimental Mathematics Lab (MXM) Spring 2025
Project: Integrating Machine Learning with Data Assimilation for Complex Dynamical System Predictions
- **Teaching Assistant** University of Wisconsin-Madison
Math 114: Precalculus Fall 2023
- **Teaching Assistant** Nanjing University
Dynamic Meteorology Fall 2021

AWARDS

- SIAM Student Travel Awards (UQ26) - 2026
- SIAM Student Travel Awards (AN25) - 2025
- UW-Madison Student Graduate Research Grants - 2025
- People's Scholarship, Nanjing University - 2017, 2018, 2019

PROGRAMMING

Python (NumPy, SciPy, pandas, PyTorch, JAX, xarray), MATLAB, R, Git, Linux